

# Documentation Log

April 7, 2026

## 1 Step 0. Read the Task and Survey Literature

Identified four deliverables: research memo, code, output datasets, documentation log. Time constraint: approximately 10 hours of working time.

**Tool:** Claude (Anthropic, web interface).

**How:** Asked Claude to act as a top empirical economist and recommend the best existing measures for patent novelty, technology uncertainty, and disclosure specificity. Claude identified Kelly et al. (2021, *AER: Insights*) as providing a pre-computed backward similarity measure available on GitHub, which saved several hours compared to constructing novelty from raw CPC citation data. Claude also recommended using the coefficient of variation of KPSS values as the uncertainty proxy, citing Pastor and Veronesi (2003) and Diether et al. (2002) as conceptual foundations.

**My contribution:** Selected which measures to use based on feasibility and literature fit. Decided to use value dispersion (CV) for uncertainty over examination duration volatility. Decided to frame disclosure specificity as an LLM-based semantic scoring task rather than shallow NLP features.

## 2 Step 1. Plan the Data Pipeline

**Tool:** Claude (web interface).

**How:** Asked Claude for exact download links, expected file sizes, merge keys, and potential data pitfalls. Claude provided GitHub URLs for KPSS and Kelly data, the PatentsView S3 download link, and warned about Git LFS requirements for large files on GitHub.

**Output:** A download checklist of three data sources (KPSS patent values, Kelly et al. novelty scores, PatentsView bibliographic data) with expected file sizes and merge key format (integer `patent_num` across KPSS/Kelly, string `patent_id` in PatentsView requiring conversion).

### 3 Step 2. Download Data and Verify

**Tool:** Git, curl, web browser.

**What happened:**

- Cloned KPSS and Kelly GitHub repos via `git clone`. Initial clone only downloaded Git LFS pointer files (few hundred bytes instead of actual data). Attempted `brew install git-lfs` but Homebrew was not installed. Resolved by downloading zip files directly from the GitHub web interface.
- PatentsView S3 link (`s3.amazonaws.com/data.patentsview.org/...`) returned Access Denied. Resolved by downloading `g_patent.tsv.zip` manually from the PatentsView website (`data.uspto.gov`).
- Verified file sizes: `g_patent.tsv` (1.11 GB), `KPSS_2024.csv` (246 MB), `Match_patent_cpc_2024.csv` (242 MB), Kelly CSV (791 MB). All reasonable.

### 4 Step 3. Write the Research Memo

**Tool:** Claude (web interface).

**How:** Provided Claude with the full task description for Parts 1, 2, and 3E. Claude produced initial drafts of each section. I then reviewed the output against the actual regression results and my own understanding of the research design, and directed multiple rounds of revision:

- I noticed the initial draft only said log transformation was due to “right skew.” I knew the percentage interpretation of coefficients was equally important, so I had Claude add that explanation.
- I recognized that CPC subclass fixed effects would absorb the uncertainty measure (since uncertainty is constructed at the subclass  $\times$  year level), and directed Claude to discuss this trade-off explicitly.
- After reviewing the regression output, I saw that the novelty coefficient was positive (+0.077), not negative as hypothesized. I analyzed why: the 2021–2022 post-COVID market may have rewarded novelty rather than discounting it. I had Claude incorporate this interpretation.
- I noticed the first-stage coefficient was negative (−0.39), which directly contradicts the mechanism chain (novelty  $\rightarrow$  higher uncertainty). I diagnosed three possible explanations myself: (a) the CV measure mechanically produces low values when patent values are uniformly low, which is likely in novel technology areas; (b) novel CPC subclasses may be internally more homogeneous; (c) text-based novelty may not align with how

investors perceive novelty. I then directed Claude to write up these explanations.

- I decided that the original 3E approach (numeric density, word length, TTR) was conceptually weak—these are surface features that miss whether a patent actually communicates its technical contribution clearly. I redesigned the approach around LLM-based scoring with a structured rubric, and had Claude write the methodology section accordingly.

**My contribution:** I performed all analysis of the regression results and identified every issue that needed to be addressed. Claude’s role was to convert my analysis into well-written academic prose. The diagnoses of the null result, the first-stage contradiction, and the redesign of the specificity measure all originated from my own reasoning.

## 5 Step 4. Generate the Analysis Code

**Tool:** Claude Code (Anthropic, terminal interface).

**How:** Wrote a `claude.md` project file specifying project background, data sources, and coding principles. Gave Claude Code a structured prompt with:

- Variable definitions (novelty = negated z-score of `bsim5`, uncertainty = CV of `xi_real` by CPC subclass  $\times$  year)
- Regression specifications (main, mechanism with interaction, first stage)
- Fixed effects (year, CPC section), clustering (CPC subclass)
- Required sanity checks (merge counts, no duplicates, standardization, control signs,  $R^2$  range)

Claude Code produced `full_analysis.py`, which:

1. Merges four datasets on `patent_num` (converting `g_patent.tsv` string IDs to integer)
2. Filters to 2021–2022 utility patents (197,446  $\rightarrow$  196,428 after merge  $\rightarrow$  194,937 after dropping small CPC groups)
3. Constructs novelty, uncertainty, and interaction variables
4. Runs three regressions with clustered standard errors
5. Exports regression tables as CSV and  $\text{\LaTeX}$
6. Includes sanity checks at every stage (all passed)

**My contribution:** Wrote the `claude.md`. Reviewed merge rates (99.5% KPSS  $\rightarrow$  Kelly, 100% thereafter). Verified control variable signs (claims +, cites +). Confirmed  $R^2$  values are plausible (0.10 main, 0.13 mechanism). No manual code edits were needed.

## 6 Step 5. Reconcile Memo and Results

After seeing the regression output:

- Updated memo with actual coefficients ( $\beta_{novelty} = 0.077$ ,  $\beta_{uncertainty} = -0.75$ ,  $\beta_{interaction} = 0.012$ ,  $\alpha_{firststage} = -0.39$ ).
- The first-stage result was negative and significant, contradicting the hypothesized mechanism. Diagnosed this as likely a mechanical artifact of the CV measure and discussed in the critical assessment.
- Null result on novelty interpreted honestly: no evidence of systematic undervaluation in 2021–2022, possibly due to post-COVID market conditions favoring novel technologies.

## 7 Step 6. Format and Package Deliverables

**Tool:** Claude (web interface), L<sup>A</sup>T<sub>E</sub>X.

**How:** Asked Claude to convert the research memo and documentation log to L<sup>A</sup>T<sub>E</sub>X. Inserted regression tables. Compiled to PDF.

**Final deliverables:**

- `research_memo.pdf` — Research memo (Parts 1, 2, 3E, references)
- `documentation_log.pdf` — This document
- `code/full_analysis.py` — Complete analysis script
- `data/merged_patent_data.csv` — Merged analysis dataset
- `output/regression_table.csv` and `.tex` — Regression results

## 8 Key Research Decisions

1. **Used Kelly et al.’s backward similarity rather than constructing a novelty measure from scratch.** Rationale: published in a top journal, covers the full sample period, publicly available. Building from raw CPC citation combinations would have taken substantially more time with unclear advantages.
2. **Constructed uncertainty at the CPC subclass  $\times$  year level.** Rationale: uncertainty is conceptually a property of the technology area, not individual patents. Aggregating also reduces direct endogeneity with the patent-level dependent variable, though using the dependent variable’s distribution remains a limitation.
3. **Used CPC section fixed effects rather than subclass fixed effects.** Rationale:

uncertainty varies at the subclass  $\times$  year level, so subclass FE would absorb it. Section-level FE is a deliberate trade-off.

4. **Proposed LLM-based scoring for disclosure specificity (3E).** Rationale: specificity is a semantic judgment requiring content understanding, not surface-level word counting. Designed a structured 1–5 rubric with validation protocol.
5. **Sample restricted to 2021–2022 as specified.** Limits statistical power and generalizability, particularly given atypical post-COVID market conditions.